

$$1-1. \lim_{x \rightarrow 0} \left[ \frac{1}{e^x - 1} - \frac{1}{\ln(1+x)} \right] = \lim_{x \rightarrow 0} \frac{\ln(1+x) - (e^x - 1)}{(e^x - 1) \ln(1+x)} = \lim_{x \rightarrow 0} \frac{[x - \frac{1}{2}x^2 + o(x^2)] - [x + \frac{1}{2}x^2 + o(x^2)]}{x^2} = -1;$$

$$1-2. \lim_{x \rightarrow 0} (2 \sin x + \cos x)^{\frac{1}{x}} = \lim_{x \rightarrow 0} (1 + 2 \sin x + \cos x - 1)^{\frac{1}{2 \sin x + \cos x - 1} \cdot \frac{2 \sin x + \cos x - 1}{x}} = e^{\lim_{x \rightarrow 0} \frac{2 \sin x + \cos x - 1}{x}} = e^2;$$

$$1-3. \lim_{x \rightarrow \infty} \left( \frac{x^2 + 1}{x + 1} - ax - b \right) = \lim_{x \rightarrow \infty} \frac{(1-a)x^2 - (a+b)x + 1-b}{x+1} = 0, \quad 1-a=0, a+b=0 \quad a=1, b=-1.$$

$$1-4. \quad x_n > 0 \quad x_{n+1} - x_n = \frac{(\sqrt{2}+1)(x_n - x_{n-1})}{(x_n + \sqrt{2})(x_{n-1} + \sqrt{2})} \quad (x_{n+1} - x_n) \quad (x_n - x_{n-1}) \quad \dots \quad (x_1 - x_0)$$

$$x_n < \sqrt{2} + 1 \quad \lim_{n \rightarrow \infty} x_n = A \quad A = \sqrt{2} + \frac{A-1}{A+\sqrt{2}}$$

$$A = \frac{1+\sqrt{5}}{2}.$$

$$1-5. \quad \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2} = \lim_{x \rightarrow 1} \frac{x+1}{x-2} = -2 \quad \lim_{x \rightarrow 2} \frac{x^2 - 1}{x^2 - 3x + 2} = \lim_{x \rightarrow 2} \frac{x+1}{x-2} = \infty \quad x=1 \quad x=2$$

$$1-6. \quad f(x) \quad x=0 \quad \lim_{x \rightarrow 0^+} x \sin \frac{1}{x} = 0 = f(0) = a \quad a=0$$

$$2-1. \quad (1) \quad (x^x)' = (e^{x \ln x})' = e^{x \ln x} \cdot (\ln x + 1) = x^x (\ln x + 1);$$

$$(2) \quad (\ln \ln \ln x)' = \frac{1}{x \ln x \ln \ln x};$$

$$(3) \quad (\ln(x + \sqrt{1+x^2}))' = \frac{1}{x + \sqrt{1+x^2}} \cdot (1 + \frac{1}{2\sqrt{1+x^2}} \cdot 2x) = \frac{1}{\sqrt{1+x^2}};$$

$$(4) \quad (f^2(\sin x))' = 2f(\sin x) \cdot f'(\sin x) \cdot \cos x;$$

$$2-2. \quad y(0) = 1 - \sin(xy) \cdot (y + xy)' - \frac{y}{x+1} \cdot \frac{y - (x+1)y'}{y^2} = 0, \quad y'(0) = 1$$

$$2-3. \quad \frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt} = \frac{1 - \frac{1}{1+t^2}}{\frac{2t}{1+t^2}} = \frac{t}{2} \quad \frac{d^2y}{dx^2} = \frac{d}{dx} \left( \frac{t}{2} \right) = \frac{d(\frac{t}{2})}{dt} \cdot \frac{1}{dx/dt} = \frac{1}{2} \cdot \frac{1+t^2}{2t} = \frac{1+t^2}{4t}$$

$$2-4. \quad y^{(20)} = x^2(e^{2x})^{(20)} + C_{20}^1 2x \cdot (e^{2x})^{(19)} + C_{20}^2 2 \cdot (e^{2x})^{(18)} = 2^{20} e^{2x} (x^2 + 20x + 95)$$

3-1.  $f(x), F(x) = x^2$  Cauchy

3-1'.  $F(x) = f(x) - x$   $F(0) = 0, F(\frac{1}{2}) = 1, F(1) = -1$   $[\frac{1}{2}, 1]$   $\exists c \in (\frac{1}{2}, 1), F(c) = 0$   $[0, c]$  Rolle

3-2.  $f'(x) = 6(x-1)(x-2)$   $x=1, x=2$   $(-\infty, 1] \cup [2, +\infty)$   $[1, 2]$

3-3.  $f''(x) = 12x(3x-2)$   $(-\infty, 0] \cup [\frac{2}{3}, +\infty)$   $[0, \frac{2}{3}]$   $(0, 1), (\frac{2}{3}, f(\frac{2}{3}))$

3-4.  $f(x) = (1+x)\ln(1+x) - \arctan x$   $f(x)$

3-5.  $f(0) = 0$   $f(\frac{2}{5}) \approx -0.33$

3-6.  $y = x - 2$

3-7.  $K = \frac{|y''|}{(1+y'^2)^{3/2}} = \frac{|2a|}{[1+(2xa+b)^2]^{3/2}} \nlessdot 2a$   $x = -\frac{b}{2a}$

4-1. 1  $\int \frac{dx}{\sqrt{1-2x}} = -\frac{1}{2} \int \frac{d(1-2x)}{\sqrt{1-2x}} = -\sqrt{1-2x} + C$

2  $\int \frac{1}{x(1+x^2)} dx = \int (\frac{1}{x} - \frac{x}{1+x^2}) dx = \ln|x| - \frac{1}{2} \ln(1+x^2) + C = \ln \frac{|x|}{\sqrt{1+x^2}} + C$

3  $\int \frac{1}{2+\sin x} dx \stackrel{\tan \frac{x}{2} = t}{=} \int \frac{dt}{t^2+t+1} = \frac{2}{\sqrt{3}} \arctan \frac{2t+1}{\sqrt{3}} + C = \frac{2}{\sqrt{3}} \arctan \frac{2 \tan \frac{x}{2} + 1}{\sqrt{3}} + C$

4  $\int \frac{\sqrt{x}}{1+\sqrt[3]{x}} dx \stackrel{\sqrt[3]{x} = t}{=} \int \frac{t^3}{1+t^2} \cdot 6t^5 dt = \int (t^6 - t^4 + t^2 - 1 + \frac{1}{1+t^2}) dt = \frac{1}{7} t^7 - \frac{1}{5} t^5 + \frac{1}{3} t^3 - t + \arctan t + C$

$= \frac{1}{7} x^{7/6} - \frac{1}{5} x^{5/6} + \frac{1}{3} x^{1/2} - x^{1/6} + \arctan x^{1/6} + C$

5  $\int x \ln x dx = \frac{1}{2} \int \ln x dx^2 = \frac{1}{2} (x^2 \ln x - \int x^2 \cdot \frac{1}{x} dx) = \frac{1}{4} x^2 (\ln x - 1) + C$

5-1.  $I = \int_0^1 x \ln(1+x) dx$

5-2.  $\lim_{x \rightarrow 0} \frac{\int_0^{\sin^2 x} \ln(1+t) dt}{\sqrt{1+x^4} - 1} = \lim_{x \rightarrow 0} \frac{\int_0^{\sin^2 x} \ln(1+t) dt}{x^4/2} = \lim_{x \rightarrow 0} \frac{\ln(1+\sin^2 x) \cdot 2 \sin x \cos x}{2x^3} = \lim_{x \rightarrow 0} \frac{\sin^2 x \cdot 2 \sin x}{2x^3} = 1$

$$5-3. \quad \int \frac{\sqrt{x}}{1+x\sqrt{x}} dx \stackrel{\sqrt{x}=t}{=} \int \frac{t}{1+t^3} \cdot 2t dt = \frac{2}{3} \ln(1+t^3) \Big|_0^2 = \frac{4}{3} \ln 3$$

$$5-4. \quad \int x \ln x dx = \frac{1}{2} (x^2 \ln x) \Big|_1^e - \int x dx = \frac{1}{2} [e^2 - \frac{1}{2}(e^2 - 1)] = \frac{1}{4}(e^2 + 1)$$

$$5-5. \quad F(x) = xf(x) \quad [0,1] \quad F(1) = f(1) = 2 \int_0^{1/2} xf(x) dx = cf(c) = F(c) \quad (0 < c < 1)$$

$$\exists \xi \in (c,1) \subset (0,1) \quad F'(\xi) = 0 \quad f'(\xi) = -\frac{f(\xi)}{\xi}.$$

$$6-1. \quad A = \int_2^4 (y + 4 - \frac{1}{2}y^2) dy = 18$$

$$6-2. \quad A = \int_0^{2\pi} \frac{1}{2} (2 + \cos \theta)^2 d\theta = \frac{9}{2} \pi$$

$$6-3. \quad s = \int_0^{2\pi} \sqrt{(x'_t)^2 + (y'_t)^2} dt = \int_0^{2\pi} \sqrt{2a^2(1 - \cos t)} dt = \int_0^{2\pi} 2a \sin \frac{t}{2} dt = 8a$$

$$V = \int_0^{2\pi a} \pi y^2 dx = \pi a^3 \int_0^{2\pi} (1 - \cos t)^3 dt = 5\pi^2 a^3$$

$$7-1. \quad \begin{array}{lll} 1 & y\sqrt{x^2+1} = C & 2 \quad \sin \frac{y}{x} = Cx \\ & & 3 \quad x^2 - xy + Cy = 0 \end{array}$$

$$4 \quad y' = \frac{y}{2(\ln y - x)} \quad \frac{dx}{dy} + \frac{2}{y}x = \frac{2 \ln y}{y} \quad x = Cy^{-2} + \ln y - \frac{1}{2}$$

$$7-2. \quad \begin{array}{ll} 1 & y = C_1 x^{-2} + C_2 x \\ 2 & \ln(\sqrt{C_1}y + \sqrt{C_1 y^2 - 1}) = \sqrt{C_1}x + C_2 \end{array}$$

$$7-3. \quad y = C_1 e^{-x} + C_2 e^{3x}$$

$$7-4. \quad Y = C_1 e^{-x} + C_2 e^x \quad y^* = \frac{x}{4}(x-1)e^x$$

$$7-5. \quad y^* = -\frac{1}{3}x \cos 2x + \frac{4}{9} \sin 2x.$$